



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Information Technology

Academic Year 2024– 2025 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT

Course Code & Title: CD3291 & Data Structures and Algorithms

Name of the Faculty member (s): Mrs. G. Mareeswari, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit II / Application of lists
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO 6,7,8
- **Activity Chosen:** Crossword Puzzle
- **Justification:**

In data structures, a **list** is a fundamental and versatile data type used to store and organize data elements in a specific order. Lists provide an efficient way to manage collections of items, offering flexibility for various operations like insertion, deletion, and traversal. Various applications of list include stack, queue, hash table also used to organize the data in an effective manner. The cross-word puzzle activity is used to check the understanding of students about the application of list concept.

- **Time Allotted for the Activity:** 10 Minutes

- **Details of the Implementation:**

- A crossword puzzle is a fun word game that encourages students to remember the idea and try to figure out the solution on their own. Instead of administering standard MCQ tests, it sparks pupils' curiosity.
- The activity was conducted in a classroom for a duration of 10 minutes. The students were given 10 questions as a crossword puzzle in the topic of application of lists. They have to think and write the answers in the given puzzle.
- The activity was conducted during the last 10 minutes of the class after that the answers were corrected. Sample answers were shown in figure 1. The answers for the wrong questions were discussed at the end of the class to make them to know the answer of all the questions.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO2	3	2	1	1	2	2

(1 – Low

2 – Moderate

3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	3	2	1	1	2	1
Justification for correlation	The students apply the knowledge of basic data structures.	The students analyze the given questions	The students find the solutions for the given question.	The students work in a team to find solutions for the given question.	The students communicate with their team members and explain the reason for providing particular solution for the question.	The students apply the knowledge of to algorithms solve the question.

• Images / Screenshot of the practice:

CROSS WORD PUZZLE
UNIT- II

Gopi. S
Vimal. S
Dharani balan. R
Sriram. T

Across

3. While try to delete element from empty queue what constrain will rise _____

6. A doubly linked list allows traversal in both _____ because each node contains pointers to both the next and the previous nodes.

8. _____ queues use circular linked lists to efficiently manage the buffer by wrapping around the end of the list back to the beginning.

9. In a _____ list implementation, each element points to only the next element in the sequence

Down

1. In an application of list insertion is performed at an end and deletion is performed at another end _____

2. While pushing element into the stack after the stack become full what constrain will rise _____

4. One of the applications of list in which both insertion and deletion can be performed at both ends _____

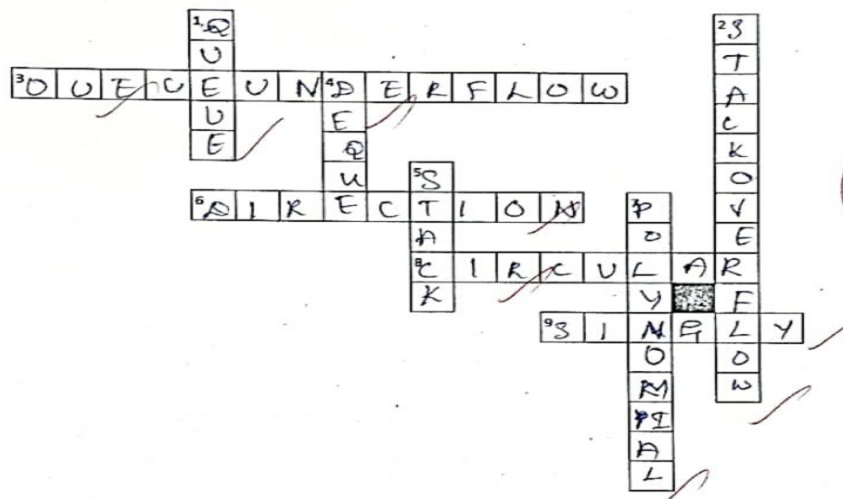
5. In an application of list insertion is done only at the top of the list _____

7. arithmetic operations can be efficiently represented using linked list

Figure 1: Sample Answers sheet shared by Gopi. S, Vimal. S, Dharanibalan. R and Sriram.T

CROSS WORD PUZZLE
UNIT- II

Jeyalakshmi. A
Ashika. A
Priyadarshini R
F. Benitha Francy



Across

3. While try to delete element from empty queue what constrain will rise _____
6. A doubly linked list allows traversal in both _____ because each node contains pointers to both the next and the previous nodes.
8. _____ queues use circular linked lists to efficiently manage the buffer by wrapping around the end of the list back to the beginning.
9. In a _____ list implementation, each element points to only the next element in the sequence

Down

1. In an application of list insertion is performed at an end and deletion is performed at another end _____
2. While pushing element into the stack after the stack become full what constrain will rise _____
4. One of the applications of list in which both insertion and deletion can be performed at both ends _____
5. In an application of list insertion is done only at the top of the list _____
7. arithmetic operations can be efficiently represented using linked list

Figure 2: Sample Answers sheet shared by Jeyalakshmi. A, Ashika. A, Priyadarshini. R, and Benitha Francy. F

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Some of the students find very easy to identify the answer for the given questions.
- Some of them take more time to complete the puzzle need more clarification on the various classes.

❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

It helps the students to think on their own and identify the uses and various applications of list in data structures, It also check different operations that involved while processing the list. The student's level of understanding on this topic gets increased.

❖ ***Challenges faced in implementation:***

- The students who didn't understand the concept clearly were unable to complete it within the allotted time.
- Few of them were struggled to find the answer for the questions.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/RV_crossword_BSEC.pdf
- ❖ <https://crosswordlabs.com/>
- ❖ https://www.ritrjpm.ac.in/images/computer-science/2022-2023/2_GM_CS3353_Pointers_CWP.pdf

Signature of Faculty Member

HOD